



## T2 Examination

**A**

**Programme: B.Tech**

**Semester: IV**

**Course Code: -**

**Course Name: DSA II**

**Branch: Computer Engineering**

**Academic Year: 2021 -2022**

**Duration: 20 mins**

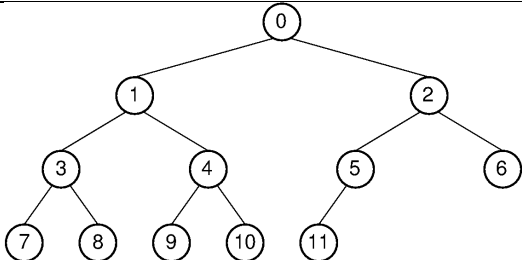
**Max Marks: 10**

**Student PRN No.**

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**Instructions:**

1. Over writing is not allowed. Such answers will not be checked.
2. No marks will be awarded for partially correct answers.

|    |                                                                                                                                                                                                                                                         |     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1. | Which of the following is/are correct preorder traversal sequence(s) of a binary search tree?<br>a. 4, 6, 7, 9, 18, 20, 19<br>b. 5, 8, 9, 12, 10, 15, 25<br>c. 3, 5, 7, 8, 15, 19, 25<br>d. 2, 7, 10, 8, 14, 16, 20<br><br><b>ANS: a, b, c, d (all)</b> | 1/2 |
| 2. | If the height of a binary tree is h. How many maximum and minimum number of nodes can it have?<br><br>Maximum: <b><math>2^{h+1} - 1</math></b><br><br>Minimum: <b><math>h + 1</math></b>                                                                | 1/2 |
| 3. | Worst Case Complexity of RL rotation of an n nodes AVL tree to balance the imbalanced node is:<br><br><b>ANS: <math>O(1)</math></b>                                                                                                                     | 1/2 |
| 4. |  <p>What is the post-order successor of node with value 4?</p> <b>ANS: 1</b>                                                                                         | 1/2 |
| 5. | Write the prototype of <code>heapify()</code> function.<br><br><b>ANS: <code>void heapify( heap *h);</code></b>                                                                                                                                         | 1/2 |
| 6. | C, B, A, D and E were inserted in an AVL tree. How many nodes will get imbalanced after the last insertion?                                                                                                                                             | 1/2 |



|     |                                                                                                                                                                                                                                                                                              |     |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|     | <b>ANS: 2</b>                                                                                                                                                                                                                                                                                |     |
| 7.  | A black node in a Red-Black Tree can have which colour children?<br><br><b>ANS: Red or black</b>                                                                                                                                                                                             | 1/2 |
| 8.  | A fully connected graph has n vertices. How many edges will it have?<br><br><b>ANS: <math>n^2</math></b>                                                                                                                                                                                     | 1/2 |
| 9.  | Adjacency Matrix of an undirected graph is:<br><br><b>ANS: SYMMETRIC</b>                                                                                                                                                                                                                     | 1/2 |
| 10. | Write the typedef for ADT heap that is implemented using array.<br><br><b>ANS:</b><br><b>typedef struct heap {</b><br><b>int *A;</b><br><b>int size;</b><br><b>int rear;</b><br><b>} heap;</b>                                                                                               | 1/2 |
| 11. | Following nodes were inserted in an AVL tree in the given order: 200, 150, 300, 180, 190, 175.<br>Now node 300 was deleted. Draw the resultant tree.<br><br><b>ANS:</b><br><b>180, 150, 200, -- , 175, 190</b><br><br>(This is level order traversal. Diagrammatic representation expected.) | 1   |
| 12. | 35, 24, 26, 23, 20, 18, 22 are the elements of a binary heap stored in an array. What is the content of the array after two consecutive delete operations?<br><br><b>ANS: 24, 23, 22, 18, 20</b>                                                                                             | 1   |
| 13. | A binary search tree is created using the integers 10, 8, 13, 11, 12 by inserting in the same order. If we want to represent this tree using array, then the node 11 will be inserted at _____ index location.<br><br><b>Ans: 5</b>                                                          | 1   |
| 14. | After deleting an element from max-heap, if maximum number of comparison operations required to re-heapify is 3. What is range of number of elements remaining in the heap now?                                                                                                              | 1   |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|     | _____ to _____                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
|     | <b>Ans: 5 to 7</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 15. | <div><pre>graph TD; 0((0)) --&gt; 1((1)); 0((0)) --&gt; 6((6)); 1((1)) --&gt; 2((2)); 2((2)) --&gt; 3((3)); 2((2)) --&gt; 4((4)); 2((2)) --&gt; 5((5)); 3((3)) --&gt; 4((4)); 4((4)) --&gt; 5((5)); 5((5)) --&gt; 2((2)); 6((6)) --&gt; 3((3));</pre></div> <div><b>ANS:</b><br/><b>0: 1-&gt;6-&gt;NULL</b><br/><b>1: 2-&gt;NULL</b><br/><b>2: 3-&gt;4-&gt;5-&gt;NULL</b><br/><b>3: 4-&gt;NULL</b><br/><b>4: 5-&gt;NULL</b><br/><b>5: NULL</b><br/><b>6: 3-&gt;NULL</b></div> | 1 |